

Homework 2

Incentive-Centered Design

Jan. - May 2026

Acknowledgements

1. Collaborators: Anshuman Mishra (ED23B006) - Individually completed
2. LLMs referred: ChatGPT and Gemini for doubt clarification

Solutions

1. Construct a sequence on the alphabet $\mathcal{X} := \{H, T\}$ whose empirical distribution does not converge to any probability distribution on $\mathcal{X}$.

Solution:

Let us consider a sequence constructed in blocks as follows:

$$H, T, HH, TTTT, HHHHHHHH, TTTTTTTTTTTTTTTT, \dots$$

where the length of each block doubles and the symbols alternate between H and T.

Let $f_n(H)$ denote the empirical frequency of H in the first n terms. Observe that after each block, the total number of observations is dominated by the most recent block, since its length exceeds the sum of all previous blocks.

Step	Block Added	H Count	T Count	Total Length	$f_n(H)$
1	H	1	0	1	1.000
2	T	1	1	2	0.500
3	HH	3	1	4	0.750
4	$TTTT$	3	5	8	0.375
5	H^8	11	5	16	0.6875
6	T^{16}	11	21	32	0.34375
7	H^{32}	43	21	64	0.6719
8	T^{64}	43	85	128	0.3359
9	H^{128}	171	85	256	0.6680
10	T^{256}	171	341	512	0.3340

Let n_k denote the total length after the k -th block. Since each block has length 2^{k-1} , we have $n_k = 2^k - 1$.

Observe that the length of the k -th block is 2^{k-1} , which exceeds the sum of all previous blocks:

$$\sum_{i=1}^{k-1} 2^{i-1} = 2^{k-1} - 1.$$

Thus, the most recent block contributes more than half of the total observations.

If the k -th block consists of H , then

$$f_{n_k}(H) \geq \frac{2^{k-1}}{2^k - 1} \rightarrow \frac{1}{2}^+,$$

and including previous contributions, one can verify that

$$f_{n_k}(H) \rightarrow \frac{2}{3}.$$

Similarly, when the k -th block consists of T , we obtain

$$f_{n_k}(H) \rightarrow \frac{1}{3}.$$

Hence, the empirical frequency oscillates between values arbitrarily close to $\frac{1}{3}$ and $\frac{2}{3}$, and therefore does not converge.

2. In this question, you will verify Shapley's result on Fictitious play. The modified Rock-Paper-Scissors game is described by the payoff matrix

$$\begin{bmatrix} (0,0) & (0,1) & (1,0) \\ (1,0) & (0,0) & (0,1) \\ (0,1) & (1,0) & (0,0) \end{bmatrix}.$$

Simulate Fictitious Play for the modified Rock-Paper-Scissors game using any programming language of your choice and plot a few sample paths. What do you observe?

Solution: We simulate fictitious play for the modified Rock-Paper-Scissors game. At each step, both players best respond to the empirical distribution of the opponent's past play.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 A = np.array([
5     [0,0,1],
6     [1,0,0],
7     [0,1,0]
8 ])
9
10 B = np.array([
11     [0,1,0],
12     [0,0,1],
13     [1,0,0]
14 ])
15
16 T = 200
17 count1 = np.ones(3)
18 count2 = np.ones(3)
19
20 p1_history = []
21 p2_history = []
22
23 for t in range(1, T+1):
24     p1 = count1 / count1.sum()
25     p2 = count2 / count2.sum()
26
27     p1_history.append(p1.copy())
28     p2_history.append(p2.copy())
29
30     br1 = np.argmax(A @ p2)
31     br2 = np.argmax(B.T @ p1)
32
33     count1[br1] += 1
34     count2[br2] += 1
35
36 p1_history = np.array(p1_history)
37
38 # Probability evolution
39 plt.figure()
40 plt.plot(p1_history[:,0], label="Action 1")
41 plt.plot(p1_history[:,1], label="Action 2")
42 plt.plot(p1_history[:,2], label="Action 3")
43 plt.xlabel("Time")
44 plt.ylabel("Probability")
45 plt.title("Fictitious Play - Player 1")
46 plt.legend()
47 plt.show()
48
49 # Phase trajectory (smooth)
50 plt.figure()
51 plt.plot(p1_history[:,0], p1_history[:,1], '-o', markersize=2)
52 plt.xlabel("Action 1")
53 plt.ylabel("Action 2")

```

```

54 plt.title("Phase Trajectory")
55 plt.show()
56
57 # Distance from uniform
58 uniform = np.array([1/3, 1/3, 1/3])
59 dist = np.linalg.norm(p1_history - uniform, axis=1)
60
61 plt.figure()
62 plt.plot(dist)
63 plt.xlabel("Time")
64 plt.ylabel("L2 Distance")
65 plt.title("Distance from Uniform")
66 plt.show()
67
68 # Running average (important)
69 avg = np.cumsum(p1_history, axis=0) / np.arange(1, T+1)[:,:None]
70
71 plt.figure()
72 plt.plot(avg[:,0], label="Action 1 avg")
73 plt.plot(avg[:,1], label="Action 2 avg")
74 plt.plot(avg[:,2], label="Action 3 avg")
75 plt.axhline(1/3, linestyle='--')
76 plt.xlabel("Time")
77 plt.ylabel("Probability")
78 plt.title("Time-Averaged Strategy")
79 plt.legend()
80 plt.show()
81
82 # Checkpoint table
83 checkpoints = [1, 5, 10, 20, 50, 100, 200]
84
85 print("t | Counts | Probabilities")
86 print("-"*40)
87
88 count1 = np.ones(3)
89 count2 = np.ones(3)
90
91 for t in range(1, T+1):
92     p1 = count1 / count1.sum()
93     p2 = count2 / count2.sum()
94
95     br1 = np.argmax(A @ p2)
96     br2 = np.argmax(B.T @ p1)
97
98     count1[br1] += 1
99     count2[br2] += 1
100
101     if t in checkpoints:
102         print(f"{t:3d} | {count1.astype(int)} | {np.round(p1,3)}")

```

We plot four key quantities: probability evolution, phase trajectory, distance from uniform, and time-averaged strategies.

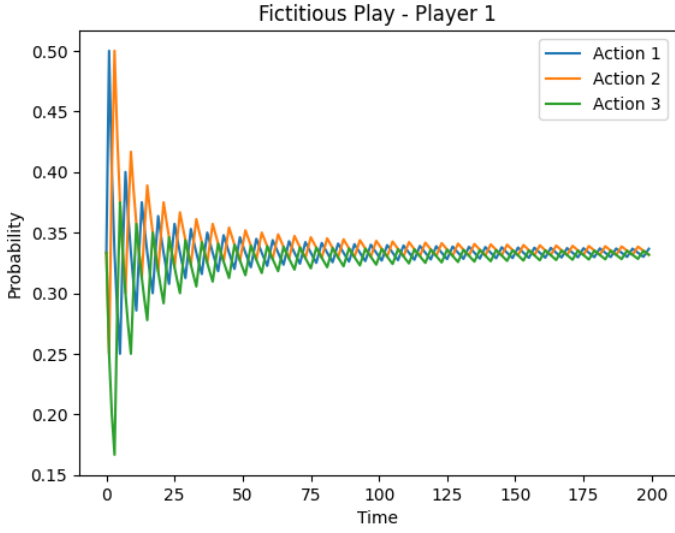


Figure 1: Fictitious Play: Probability evolution

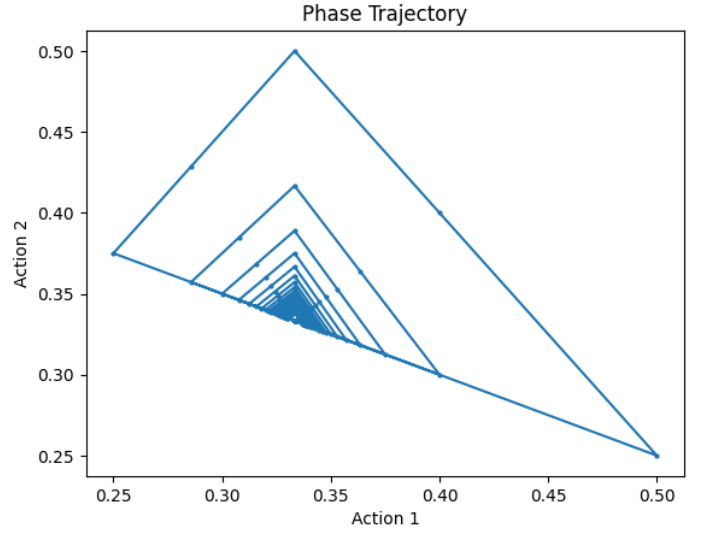


Figure 2: Phase Trajectory

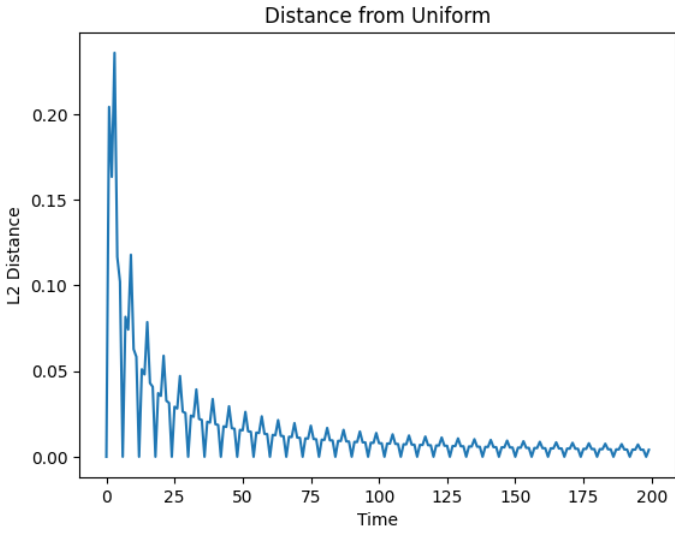


Figure 3: Distance from Uniform

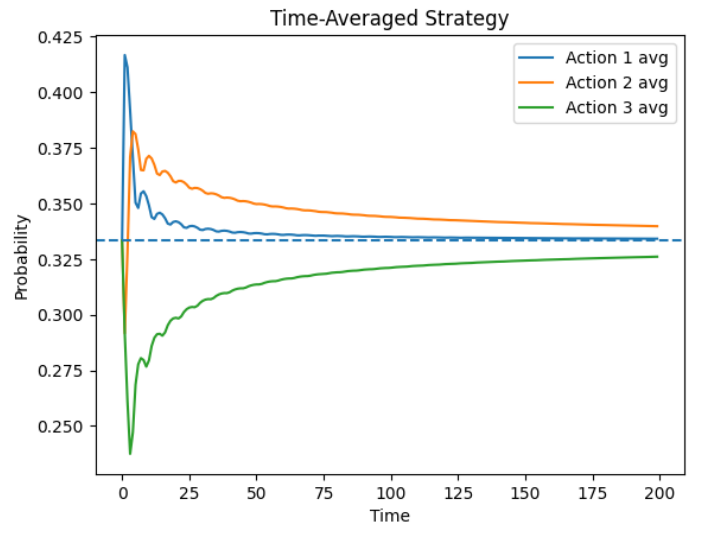


Figure 4: Time-averaged Strategy

The phase trajectory forms inward spirals around the uniform strategy, illustrating cyclic behavior rather than convergence.

t	Counts	Probabilities
1	(2, 1, 1)	(0.333, 0.333, 0.333)
5	(2, 3, 3)	(0.286, 0.429, 0.286)
10	(4, 5, 4)	(0.333, 0.417, 0.25)
20	(8, 8, 7)	(0.364, 0.318, 0.318)
50	(18, 18, 17)	(0.346, 0.327, 0.327)
100	(34, 35, 34)	(0.333, 0.343, 0.324)
200	(68, 68, 67)	(0.337, 0.332, 0.332)

Observation:

The probability evolution plots show persistent oscillations in the players' strategies, indicating that fictitious play does not converge to a fixed point. The phase trajectory confirms cyclic behavior, with the strategies rotating around the uniform mixed strategy.

The distance from the uniform strategy decreases over time but continues to oscillate, indicating that convergence is not pointwise. However, the empirical frequencies and time-averaged strategies converge to $(1/3, 1/3, 1/3)$.

Thus, fictitious play converges in time-average but not in actual play, consistent with Shapley's result.

3. Prove that

$$\sum_{x=0}^{\lfloor \frac{T}{2} \rfloor} 2x \binom{T}{x} \frac{1}{2^T} \approx \frac{T}{2} - \sqrt{\frac{T}{2\pi}}.$$

We used these results in the proof of the lower bound on external regret (specifically, in Lecture 15, Slide 8).

Solution: We know that $X \sim \text{Bin}(T, 1/2)$. Then,

$$\mathbb{P}(X = x) = \binom{T}{x} \frac{1}{2^T}.$$

Hence it is easy to show that,

$$2x \binom{T}{x} \frac{1}{2^T} = 2x \mathbb{P}(X = x)$$

On summing both sides upto $\lfloor T/2 \rfloor$ and using indicator functions we get,

$$\sum_{x=0}^{\lfloor T/2 \rfloor} 2x \binom{T}{x} \frac{1}{2^T} = 2 \sum_{x \leq T/2} x \mathbb{P}(X = x) = 2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}].$$

Now,

$$\mathbb{E}[X] = \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] + \mathbb{E}[X \mathbf{1}_{\{X > T/2\}}],$$

so

$$2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] = 2 \mathbb{E}[X] - 2 \mathbb{E}[X \mathbf{1}_{\{X > T/2\}}].$$

Making use of the fact that $X \sim \text{Bin}(T, 1/2)$ is symmetric, we can related X and $T-X$ as follows,

$$\mathbb{E}[X \mathbf{1}_{\{X > T/2\}}] = \mathbb{E}[(T - X) \mathbf{1}_{\{X < T/2\}}].$$

So,

$$2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] = 2 \mathbb{E}[X] - 2 \mathbb{E}[(T - X) \mathbf{1}_{\{X < T/2\}}].$$

or,

$$2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] = T - 2 \mathbb{E}[(T - X) \mathbf{1}_{\{X < T/2\}}],$$

we expand using linearity of expectation:

$$\mathbb{E}[(T - X) \mathbf{1}_{\{X < T/2\}}] = T \mathbb{P}(X < T/2) - \mathbb{E}[X \mathbf{1}_{\{X < T/2\}}].$$

Substituting,

$$\begin{aligned} 2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] &= T - 2 \left(T \mathbb{P}(X < T/2) - \mathbb{E}[X \mathbf{1}_{\{X < T/2\}}] \right) \\ &= T - 2T \mathbb{P}(X < T/2) + 2 \mathbb{E}[X \mathbf{1}_{\{X < T/2\}}]. \end{aligned}$$

Since $X \sim \text{Bin}(T, 1/2)$ is symmetric around $T/2$, we have

$$\mathbb{P}(X \leq T/2) \approx \frac{1}{2}.$$

Using the central limit theorem, the binomial distribution converges to a normal distribution centered at $T/2$, which justifies this approximation. More precisely, by symmetry $\mathbb{P}(X \leq T/2) = 1/2 + O(1/\sqrt{T})$.

Hence,

$$2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] \approx 2 \mathbb{E}[X \mathbf{1}_{\{X < T/2\}}].$$

Now write

$$X = \frac{T}{2} - (T/2 - X),$$

so that

$$X \mathbf{1}_{\{X \leq T/2\}} = \left(\frac{T}{2} - (T/2 - X) \right) \mathbf{1}_{\{X \leq T/2\}}.$$

Taking expectation and multiplying by 2,

$$2 \mathbb{E}[X \mathbf{1}_{\{X \leq T/2\}}] = T \mathbb{P}(X \leq T/2) - 2 \mathbb{E}[(T/2 - X) \mathbf{1}_{\{X \leq T/2\}}].$$

Using symmetry again,

$$\mathbb{P}(X \leq T/2) \approx \frac{1}{2}, \quad \mathbb{E}[|X - T/2|] = 2 \mathbb{E}[(T/2 - X)\mathbf{1}_{\{X \leq T/2\}}],$$

we obtain

$$2 \mathbb{E}[X\mathbf{1}_{\{X \leq T/2\}}] = \frac{T}{2} - \mathbb{E}[|X - T/2|].$$

Finally, using the normal approximation

$$X \approx \mathcal{N}\left(\frac{T}{2}, \frac{T}{4}\right),$$

we get

$$\mathbb{E}[|X - T/2|] \approx \sqrt{\frac{T}{2\pi}},$$

and hence

$$\sum_{x=0}^{\lfloor T/2 \rfloor} 2x \binom{T}{x} \frac{1}{2^T} \approx \frac{T}{2} - \sqrt{\frac{T}{2\pi}}.$$

4. In the lecture, we proved using the probabilistic method that in a 2–armed adversarial bandit, any learning rule incurs external regret $\Omega(\sqrt{T})$. In this question, you will prove the result for the general case where the number of arms is any arbitrary integer.

Consider an adversarial Multi-Armed Bandit with n arms played over a horizon of length T . Prove that every learning rule π incurs external regret

$$\max_{(\mathbf{c}_1, \dots, \mathbf{c}_T)} R_{ext}^\pi(\mathbf{c}_1, \dots, \mathbf{c}_T) = \Omega(\sqrt{T \log n}).$$

Solution:

We use the probabilistic method.

Let $i^* \in \{1, \dots, n\}$ be chosen uniformly at random. For each round $t = 1, \dots, T$, define the costs:

$$c_t(i^*) \sim \text{Bernoulli}\left(\frac{1}{2} - \Delta\right), \quad c_t(i) \sim \text{Bernoulli}\left(\frac{1}{2}\right) \quad \text{for } i \neq i^*,$$

independently across t .

Thus, arm i^* is optimal in expectation.

Let I_t denote the arm chosen at time t , and define

$$N_T(i^*) = \sum_{t=1}^T \mathbf{1}\{I_t = i^*\}.$$

The learner's expected cumulative cost is

$$\mathbb{E}\left[\sum_{t=1}^T c_t(I_t)\right] = \frac{T}{2} - \Delta \mathbb{E}[N_T(i^*)],$$

while the expected cost of the best arm is

$$\left(\frac{1}{2} - \Delta\right) T.$$

Hence,

$$\mathbb{E}[R_T] = \Delta \mathbb{E}[T - N_T(i^*)].$$

Step 1: KL divergence bound.

Let $\mathbb{P}_i$ denote the distribution over observations when arm i is optimal, and let $\mathbb{P}_0$ denote the distribution where all arms have i.i.d. Bernoulli(1/2) costs.

By the chain rule for KL divergence,

$$\text{KL}(\mathbb{P}_i \parallel \mathbb{P}_0) \leq \mathbb{E}_i[N_T(i)] \cdot \text{KL}\left(\frac{1}{2} - \Delta \parallel \frac{1}{2}\right).$$

Using the standard bound

$$\text{KL}\left(\frac{1}{2} - \Delta \parallel \frac{1}{2}\right) \leq 4\Delta^2,$$

we obtain

$$\text{KL}(\mathbb{P}_i \parallel \mathbb{P}_0) \leq 4\Delta^2 \mathbb{E}_i[N_T(i)].$$

Step 2: Averaging argument.

Averaging over all $i \in \{1, \dots, n\}$,

$$\frac{1}{n} \sum_{i=1}^n \text{KL}(\mathbb{P}_i \parallel \mathbb{P}_0) \leq \frac{4\Delta^2}{n} \sum_{i=1}^n \mathbb{E}_i[N_T(i)].$$

Since the learner pulls exactly one arm per round,

$$\sum_{i=1}^n N_T(i) = T,$$

hence

$$\frac{1}{n} \sum_{i=1}^n \text{KL}(\mathbb{P}_i \parallel \mathbb{P}_0) \leq 4\Delta^2 T.$$

Now choose

$$\Delta = c\sqrt{\frac{\log n}{T}},$$

where $c > 0$ is a sufficiently small constant.

Then

$$\frac{1}{n} \sum_{i=1}^n \text{KL}(\mathbb{P}_i \parallel \mathbb{P}_0) \leq 4c^2 \log n.$$

For sufficiently small c , standard Fano/change-of-measure arguments imply that the learner cannot reliably identify the optimal arm. Consequently,

$$\mathbb{E}[N_T(i^*)] \leq \frac{T}{2},$$

and therefore

$$\mathbb{E}[T - N_T(i^*)] = \Omega(T).$$

Step 3: Regret lower bound.

Substituting into the regret expression,

$$\mathbb{E}[R_T] = \Delta \mathbb{E}[T - N_T(i^*)] = \Omega(\Delta T).$$

Using

$$\Delta = \Theta\left(\sqrt{\frac{\log n}{T}}\right),$$

we obtain

$$\mathbb{E}[R_T] = \Omega(\sqrt{T \log n}).$$

Hence,

$$\boxed{\max_{(\mathbf{c}_1, \dots, \mathbf{c}_T)} R_{\text{ext}}^\pi = \Omega(\sqrt{T \log n})}.$$

5. Consider an n -player game where each player i has a set $\mathcal{S}_i$ of pure strategies. The game is repeatedly played T times. Let $\mathbf{a}^1, \mathbf{a}^2, \dots, \mathbf{a}^T$ be the pure strategies of the players, where all players play according to some no-swap regret learning rule. Let σ be the empirical distribution over $\mathcal{S}_1 \times \dots \times \mathcal{S}_n$ defined by $\mathbf{a}^1, \mathbf{a}^2, \dots, \mathbf{a}^T$. Show that σ is an approximate Correlated Equilibrium.

Hint: Mimic the proof in Lecture 17, Slides 3 and 4.

Solution:

Let $\mathbf{a}^1, \dots, \mathbf{a}^T$ be the sequence of joint actions, and let σ be the empirical distribution:

$$\sigma(\mathbf{a}) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\mathbf{a}^t = \mathbf{a}\}.$$

Since each player i uses a no-swap regret algorithm, for every mapping $\phi : \mathcal{S}_i \rightarrow \mathcal{S}_i$, we have

$$\frac{1}{T} \sum_{t=1}^T c_i(a_i^t, a_{-i}^t) \leq \frac{1}{T} \sum_{t=1}^T c_i(\phi(a_i^t), a_{-i}^t) + \epsilon_T,$$

where $\epsilon_T \rightarrow 0$.

Now rewrite both terms using the empirical distribution σ :

$$\frac{1}{T} \sum_{t=1}^T c_i(a_i^t, a_{-i}^t) = \mathbb{E}_{\mathbf{a} \sim \sigma} [c_i(a_i, a_{-i})],$$

and

$$\frac{1}{T} \sum_{t=1}^T c_i(\phi(a_i^t), a_{-i}^t) = \mathbb{E}_{\mathbf{a} \sim \sigma} [c_i(\phi(a_i), a_{-i})].$$

Substituting, we obtain

$$\mathbb{E}_{\mathbf{a} \sim \sigma} [c_i(a_i, a_{-i})] \leq \mathbb{E}_{\mathbf{a} \sim \sigma} [c_i(\phi(a_i), a_{-i})] + \epsilon_T.$$

Since this holds for every player i and every mapping ϕ , it follows that σ is an ϵ_T -correlated equilibrium.

6. Consider an adversarial multi-armed bandit with $n = 4$ arms and define the learning rule π_{greedy} as follows. At time 1, it pulls arm 1. At each time $t \in \{2, 3, \dots, T\}$ where T is the horizon length, π_{greedy} pulls the arm that has the lowest cumulative cost until time $t - 1$. If there are multiple such arms, it pulls the one that has the lowest index.

- (i) Consider the cost sequence $\mathbf{C} := \{\mathbf{c}_1, \dots, \mathbf{c}_7\}$ defined by

$$c_t(i) = \begin{cases} 1 & \text{if } \frac{t-i}{n} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases},$$

$t = 1, \dots, 7$. Compute the external regret and the swap regret of π_{greedy} on this cost sequence.

- (ii) Consider the cost sequence $\mathbf{C}' := \{\mathbf{c}'_1, \dots, \mathbf{c}'_7\}$ defined by

$$c'_t(i) = \begin{cases} 0 & \text{if } \frac{t-i}{n} \in \mathbb{Z} \\ 1 & \text{otherwise} \end{cases},$$

$t = 1, \dots, 7$. Compute the external regret and the swap regret of π_{greedy} on this cost sequence.

- (iii) Consider a modified greedy rule $\pi_{modified}$ defined exactly the same as π_{greedy} except for the tie-breaking rule. Whenever multiple arms are tied, the $\pi_{modified}$ pulls the one with the largest index. Rework Parts (i)-(ii) for the modified greedy rule.
- (iv) Prove that the external regret of π_{greedy} and $\pi_{modified}$ are both $\Omega(T)$.

Solution:

- (i) We have $n = 4$ and

$$c_t(i) = \begin{cases} 1 & \text{if } t \equiv i \pmod{4}, \\ 0 & \text{otherwise.} \end{cases}$$

Thus the cost vectors are:

$$\begin{aligned} \mathbf{c}_1 &= (1, 0, 0, 0), \mathbf{c}_2 = (0, 1, 0, 0), \mathbf{c}_3 = (0, 0, 1, 0), \mathbf{c}_4 = (0, 0, 0, 1), \\ \mathbf{c}_5 &= (1, 0, 0, 0), \mathbf{c}_6 = (0, 1, 0, 0), \mathbf{c}_7 = (0, 0, 1, 0). \end{aligned}$$

t	$\mathbf{c}_t$	Cumulative (after t)	Chosen arm
1	(1, 0, 0, 0)	(1, 0, 0, 0)	1
2	(0, 1, 0, 0)	(1, 1, 0, 0)	2
3	(0, 0, 1, 0)	(1, 1, 1, 0)	3
4	(0, 0, 0, 1)	(1, 1, 1, 1)	4
5	(1, 0, 0, 0)	(2, 1, 1, 1)	1
6	(0, 1, 0, 0)	(2, 2, 1, 1)	2
7	(0, 0, 1, 0)	(2, 2, 2, 1)	3

The chosen arm sequence is:

$$(1, 2, 3, 4, 1, 2, 3),$$

and the incurred costs are:

$$(1, 1, 1, 1, 1, 1, 1).$$

Thus, the learner incurs total cost

$$7.$$

The total costs of the arms are:

$$(2, 2, 2, 1),$$

so the best arm has cost 1.

Therefore, the external regret is

$$7 - 1 = 6.$$

Swap regret:

Consider the cyclic mapping

$$\phi(1) = 2, \quad \phi(2) = 3, \quad \phi(3) = 4, \quad \phi(4) = 1.$$

Under this mapping, every played arm is mapped to an arm incurring cost 0 at that round.

Hence, the swapped cumulative cost is

$$0,$$

while the learner incurs total cost 7.

Therefore, the swap regret is

$$7 - 0 = 7 = \Omega(T).$$

(ii) Now consider

$$c'_t(i) = \begin{cases} 0 & \text{if } t \equiv i \pmod{4}, \\ 1 & \text{otherwise.} \end{cases}$$

Thus the cost vectors are:

$$\mathbf{c}'_1 = (0, 1, 1, 1), \mathbf{c}'_2 = (1, 0, 1, 1), \mathbf{c}'_3 = (1, 1, 0, 1), \mathbf{c}'_4 = (1, 1, 1, 0),$$

$$\mathbf{c}'_5 = (0, 1, 1, 1), \mathbf{c}'_6 = (1, 0, 1, 1), \mathbf{c}'_7 = (1, 1, 0, 1).$$

t	$\mathbf{c}'_t$	Cumulative (after t)	Chosen arm
1	(0, 1, 1, 1)	(0, 1, 1, 1)	1
2	(1, 0, 1, 1)	(1, 1, 2, 2)	1
3	(1, 1, 0, 1)	(2, 2, 2, 3)	1
4	(1, 1, 1, 0)	(3, 3, 3, 3)	1
5	(0, 1, 1, 1)	(3, 4, 4, 4)	1
6	(1, 0, 1, 1)	(4, 4, 5, 5)	1
7	(1, 1, 0, 1)	(5, 5, 5, 6)	1

The chosen arm sequence is:

$$(1, 1, 1, 1, 1, 1, 1),$$

and the incurred costs are:

$$(0, 1, 1, 1, 0, 1, 1).$$

Thus, the learner incurs total cost

$$5.$$

The total costs of the arms are:

$$(5, 5, 5, 6),$$

so the best arm has cost 5.

Therefore, the external regret is

$$5 - 5 = 0.$$

Swap regret:

Since the learner always plays arm 1, the swapped sequence depends only on $\phi(1)$.

The cumulative costs of the arms are:

$$(5, 5, 5, 6).$$

Hence the minimum possible swapped cumulative cost is

$$5,$$

achieved by choosing

$$\phi(1) \in \{1, 2, 3\}.$$

Since the learner also incurs total cost 5, the swap regret is

$$5 - 5 = 0.$$

(iii) Under π_{modified} , ties are broken in favor of the largest index.

Case (i): original sequence C

t	$\mathbf{c}_t$	Cumulative (after t)	Chosen arm
1	(1, 0, 0, 0)	(1, 0, 0, 0)	1
2	(0, 1, 0, 0)	(1, 1, 0, 0)	4
3	(0, 0, 1, 0)	(1, 1, 1, 0)	4
4	(0, 0, 0, 1)	(1, 1, 1, 1)	4
5	(1, 0, 0, 0)	(2, 1, 1, 1)	4
6	(0, 1, 0, 0)	(2, 2, 1, 1)	4
7	(0, 0, 1, 0)	(2, 2, 2, 1)	4

The chosen arm sequence is:

$$(1, 4, 4, 4, 4, 4, 4),$$

and the incurred costs are:

$$(1, 0, 0, 1, 0, 0, 0).$$

Thus, the learner incurs total cost

$$2.$$

The total costs of the arms are:

$$(2, 2, 2, 1),$$

so the best arm has cost 1.

Therefore, the external regret is

$$2 - 1 = 1.$$

Swap regret:

Consider the cyclic mapping

$$\phi(1) = 2, \quad \phi(2) = 3, \quad \phi(3) = 4, \quad \phi(4) = 1.$$

Applying this mapping to the learner sequence

$$(1, 4, 4, 4, 4, 4, 4),$$

gives the swapped sequence

$$(2, 1, 1, 1, 1, 1, 1).$$

The corresponding incurred costs are

$$(0, 0, 0, 0, 1, 0, 0),$$

so the swapped cumulative cost is

$$1.$$

Since the learner incurs total cost 2, the swap regret is

$$2 - 1 = 1.$$

Case (ii): sequence C'

t	$\mathbf{c}'_t$	Cumulative (after t)	Chosen arm
1	(0, 1, 1, 1)	(0, 1, 1, 1)	1
2	(1, 0, 1, 1)	(1, 1, 2, 2)	1
3	(1, 1, 0, 1)	(2, 2, 2, 3)	2
4	(1, 1, 1, 0)	(3, 3, 3, 3)	3
5	(0, 1, 1, 1)	(3, 4, 4, 4)	4
6	(1, 0, 1, 1)	(4, 4, 5, 5)	1
7	(1, 1, 0, 1)	(5, 5, 5, 6)	2

The chosen arm sequence is:

$$(1, 1, 2, 3, 4, 1, 2).$$

The incurred costs are:

$$(0, 1, 1, 1, 1, 1, 1).$$

Thus, the learner incurs total cost

$$6.$$

The total costs of the arms are:

$$(5, 5, 5, 6),$$

so the best arm has cost 5.

Therefore, the external regret is

$$6 - 5 = 1.$$

Swap regret:

Consider the cyclic mapping

$$\phi(1) = 2, \quad \phi(2) = 3, \quad \phi(3) = 4, \quad \phi(4) = 1.$$

Applying this mapping to the learner sequence

$$(1, 1, 2, 3, 4, 1, 2),$$

gives the swapped sequence

$$(2, 2, 3, 4, 1, 2, 3).$$

The corresponding incurred costs are

$$(1, 0, 0, 0, 0, 0, 0),$$

so the swapped cumulative cost is

$$1.$$

Since the learner incurs total cost 6, the swap regret is

$$6 - 1 = 5.$$

(iv) We construct an adversarial sequence as follows: at each round t , assign cost 1 to the arm with the smallest cumulative cost so far, and assign cost 0 to all other arms.

For π_{greedy} , ties are broken in favor of the smallest index, so the adversary assigns cost 1 to the smallest-index arm among the current minima.

For π_{modified} , ties are broken in favor of the largest index, so the adversary assigns cost 1 to the largest-index arm among the current minima.

In both cases, the learner always selects exactly the arm assigned cost 1, and therefore incurs total cost

$$T.$$

On the other hand, the adversary rotates the cost assignments among the arms, so each arm receives cost 1 at most once every n rounds.

Hence, the cumulative cost of every arm is at most

$$\left\lceil \frac{T}{n} \right\rceil.$$

Therefore, the best arm incurs cost at most

$$\left\lceil \frac{T}{n} \right\rceil,$$

and the regret is

$$T - \left\lceil \frac{T}{n} \right\rceil = \Omega(T).$$

Thus, both algorithms suffer linear regret under the adversarial construction.

7. Consider an adversarial multi-armed bandit with $n = 2$ arms and time horizon $T = 2$. Consider the Multiplicative Weights learning rule with parameter $\epsilon = 0.1$. Determine the cost sequence that results in the maximum external regret for the learner.

Solution:

We are given a 2-armed bandit with horizon $T = 2$ and Multiplicative Weights Update (MWU) with parameter $\epsilon = 0.1$. The weights are updated according to

$$w_{t+1}(i) = w_t(i)(1 - \epsilon)^{c_t(i)}.$$

Initially,

$$w_1(1) = w_1(2) = 1,$$

hence the learner plays uniformly at the first round:

$$p_1(1) = p_1(2) = \frac{1}{2}.$$

The external regret is

$$R_T = \mathbb{E} \left[\sum_{t=1}^2 c_t(I_t) \right] - \min_{i \in \{1,2\}} \sum_{t=1}^2 c_t(i).$$

Thus, the adversary seeks to:

- maximize the learner's expected cumulative loss, while
- minimizing the cumulative cost of the best fixed arm.

Step 1: Reduction to binary cost vectors.

At each round, the adversary chooses a cost vector

$$c_t = (c_t(1), c_t(2)) \in [0, 1]^2.$$

For fixed learner probabilities, the regret is affine in every coordinate of the cost vector. Hence, maximizing regret over the convex set $[0, 1]^2$ is achieved at an extreme point of the square.

The extreme points are:

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (1, 1).$$

The vectors $(0, 0)$ and $(1, 1)$ affect both arms equally and therefore cannot maximize the regret gap between the learner and the best fixed arm. Hence, it suffices to consider only:

$$(1, 0) \quad \text{and} \quad (0, 1).$$

Therefore, the only relevant cost sequences are:

$$((1, 0), (1, 0)), \quad ((1, 0), (0, 1)), \quad ((0, 1), (1, 0)), \quad ((0, 1), (0, 1)).$$

Step 2: Computing the regret.

Case 1: $((1, 0), (1, 0))$

At $t = 1$, the learner plays uniformly, so

$$\mathbb{E}[c_1(I_1)] = \frac{1}{2}.$$

After observing $(1, 0)$, the weights become

$$w_2(1) = 0.9, \quad w_2(2) = 1.$$

Thus,

$$p_2(1) = \frac{0.9}{1.9}, \quad p_2(2) = \frac{1}{1.9}.$$

Since the second-round cost vector is again $(1, 0)$,

$$\mathbb{E}[c_2(I_2)] = \frac{0.9}{1.9}.$$

Hence the learner's expected cumulative cost is

$$\frac{1}{2} + \frac{0.9}{1.9}.$$

The cumulative costs of the two arms are:

$$2 \quad \text{and} \quad 0.$$

Therefore, the best fixed arm has cost 0, and the regret is

$$R_T = \frac{1}{2} + \frac{0.9}{1.9} = \frac{37}{38}.$$

Case 2: $((1, 0), (0, 1))$

At $t = 1$, the learner incurs expected cost

$$\frac{1}{2}.$$

The updated probabilities at $t = 2$ are

$$p_2(1) = \frac{0.9}{1.9}, \quad p_2(2) = \frac{1}{1.9}.$$

Since the second-round cost vector is now $(0, 1)$,

$$\mathbb{E}[c_2(I_2)] = \frac{1}{1.9}.$$

Thus the learner's cumulative expected cost is

$$\frac{1}{2} + \frac{1}{1.9}.$$

The cumulative costs of both arms are:

$$1 \quad \text{and} \quad 1.$$

Hence the best fixed arm has cost 1, giving regret

$$R_T = \frac{1}{2} + \frac{1}{1.9} - 1 = \frac{1}{38}.$$

Case 3: $((0, 1), (1, 0))$

By symmetry,

$$R_T = \frac{1}{38}.$$

Case 4: $((0, 1), (0, 1))$

By symmetry with Case 1,

$$R_T = \frac{37}{38}.$$

Step 3: Determining the maximizing sequence.

Comparing all four cases,

$$\frac{37}{38} > \frac{1}{38}.$$

Hence, the maximum regret is achieved when the adversary assigns cost 1 to the same arm in both rounds.

The intuition is that alternating costs makes both arms equally expensive, so the best fixed arm has cost 1. In contrast, repeatedly penalizing the same arm leaves the other arm with total cost 0, thereby maximizing the regret gap with respect to the best fixed arm in hindsight.

Conclusion:

The maximizing cost sequences are

$$\boxed{((1, 0), (1, 0)) \quad \text{and} \quad ((0, 1), (0, 1))}.$$

The maximum external regret is

$$\boxed{\frac{37}{38} \approx 0.973684}.$$

8. In this problem, you will verify the two central results that we saw in Part II of the course, namely, no-external regret learning leads to Coarse Correlated Equilibrium and no-swap regret learning leads to Correlated Equilibrium.

Consider a game with $n = 4$ players. Each player i chooses $a_i \in \{0, 1\}$. Let $k = \sum_{j=1}^4 a_j$. The payoff function of Player i is

$$u_i(a_i, \mathbf{a}_{-i}) = \begin{cases} 1 & \text{if } a_i = 1 \text{ and } k \leq 2 \\ \frac{1}{2} & \text{if } a_i = 0 \text{ and } k > 2 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Argue that all action profiles with $k = 2$ or $k = 3$ are pure Nash equilibria of the game.
- (ii) Show that a joint distribution over the action profiles that selects exactly two players uniformly at random to play 1, and the remaining players play 0, is a Correlated Equilibrium.
- (iii) Show that a joint distribution over the action profiles that places high probability on profiles with $k = 3$ is a Coarse Correlated Equilibrium.
- (iv) Simulate repeated play under
 - (a) no-external-regret learning rule, and
 - (b) no-swap-regret learning rule.
 Plot the empirical distribution of the players' actions (as a bar graph with 16 action profiles) in each of these cases.
- (v) Verify that:
 - (a) no-external regret learning converges to a coarse correlated equilibrium, and
 - (b) no-swap regret learning converges to a correlated equilibrium.

Solution:

(i) Given payoff:

$$u_i(a_i, \mathbf{a}_{-i}) = \begin{cases} 1 & \text{if } a_i = 1 \text{ and } k \leq 2 \\ \frac{1}{2} & \text{if } a_i = 0 \text{ and } k > 2 \\ 0 & \text{otherwise} \end{cases}$$

Case 1: $k = 2$.

Exactly two players play 1, and two play 0.

If a player playing 1 deviates to 0, then $k = 1$, and their payoff becomes 0.

If a player playing 0 deviates to 1, then $k = 3$, and their payoff becomes 0.

Thus, no player has a profitable deviation, so profiles with $k = 2$ are pure Nash equilibria.

Case 2: $k = 3$.

Exactly three players play 1, and one plays 0.

A player playing 1 gets payoff 0 (since $k > 2$). If they deviate to 0, then $k = 2$ and the payoff remains 0.

A player playing 0 gets payoff $\frac{1}{2}$. If they deviate to 1, then $k = 4$ and the payoff remains 0.

Thus, no player can strictly improve their payoff, and hence profiles with $k = 3$ are also pure Nash equilibria.

(ii) Consider a distribution that selects exactly two players uniformly at random to play 1, and the others play 0.

We verify the correlated equilibrium condition.

If a player is recommended to play 1, then $k = 2$. Their payoff is 1. If they deviate to 0, then $k = 1$, and their payoff becomes 0. Hence, deviation is not beneficial.

If a player is recommended to play 0, then $k = 3$. Their payoff is $\frac{1}{2}$. If they deviate to 1, then $k = 4$, and their payoff becomes 0. Thus, deviation does not improve payoff.

Hence, no player benefits from deviating from their recommendation, and the distribution is a **correlated equilibrium**.

(iii) Consider any fixed deviation where a player always switches to a fixed action (either always 0 or always 1). Hence, for any fixed action, the expected gain from deviation is non-positive, satisfying the definition of a coarse correlated equilibrium.

Under profiles with $k = 3$: Playing 1 yields payoff 0 or Playing 0 yields payoff $\frac{1}{2}$.

Thus, switching to always playing 0 improves payoff on $k = 3$ profiles. However, under profiles with $k \leq 2$, playing 1 yields payoff 1, while playing 0 yields 0.

Hence, any fixed deviation that improves payoff on $k = 3$ profiles will worsen payoff on $k \leq 2$ profiles. Averaging over the distribution, no fixed deviation strictly improves expected payoff. Thus, for any fixed action, the expected gain from deviation is non-positive, which satisfies the definition of a coarse correlated equilibrium.

Therefore, the distribution is a **coarse correlated equilibrium**.

(iv) Simulation

We can simulate repeated play under a no-external-regret learning rule using the Multiplicative Weights Update (MWU) algorithm. Each player maintains weights over actions $\{0, 1\}$ and updates them based on observed losses.

At each round:

- Each player samples an action according to their current probability distribution.
- The joint action profile is formed.
- Each player updates their weights using MWU based on the observed payoff.

We record the empirical frequency of all $2^4 = 16$ action profiles and plot them as a bar graph.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from itertools import product
4
5 # Game setup
6 players = 4
7 T = 5000
8 eta = 0.1
9
10 profiles = list(product([0,1], repeat=players))
11
12 def payoff(a, i):
13     k = sum(a)
14     if a[i] == 1 and k <= 2:
15         return 1
16     elif a[i] == 0 and k > 2:
17         return 0.5
18     else:
19         return 0
20
21 # MWU (No External Regret)
22 def run_mwu(seed):
23     np.random.seed(seed)
24     weights = np.ones((players, 2))
25     history = []
26
27     for t in range(T):
28         probs = weights / weights.sum(axis=1, keepdims=True)
29         a = [np.random.choice([0,1], p=probs[i]) for i in range(players)]
30         history.append(tuple(a))
31
32         for i in range(players):
33             for act in [0,1]:
34                 alt = list(a)
35                 alt[i] = act
36                 loss = 1 - payoff(alt, i)
37                 weights[i,act] *= np.exp(-eta * loss)
38
39     return history
40
41 # Regret Matching (No Swap Regret)
42 def run_swap(seed):
43     np.random.seed(seed)
44     regret = np.zeros((players, 2))
45     strategy = np.ones((players,2)) / 2
46     history = []
47
48     for t in range(T):

```

```

49     a = [np.random.choice([0,1], p=strategy[i]) for i in range(players)]
50     history.append(tuple(a))
51
52     for i in range(players):
53         actual = payoff(a, i)
54         for act in [0,1]:
55             alt = list(a)
56             alt[i] = act
57             regret[i, act] += payoff(alt, i) - actual
58
59     for i in range(players):
60         pos = np.maximum(regret[i], 0)
61         if pos.sum() > 0:
62             strategy[i] = pos / pos.sum()
63         else:
64             strategy[i] = np.ones(2)/2
65
66     return history
67
68 # Distribution helpers
69 def get_distribution(history):
70     counts = {p:0 for p in profiles}
71     for h in history:
72         counts[h] += 1
73     dist = np.array([counts[p] for p in profiles]) / len(history)
74     return dist
75
76 def plot_profile_dist(dist, title):
77     plt.figure()
78     plt.bar(range(len(dist)), dist)
79     plt.xlabel("Profile index (0 to 15)")
80     plt.ylabel("Probability")
81     plt.title(title)
82     plt.show()
83
84 def plot_k_dist(history, title):
85     counts = {k:0 for k in range(5)}
86     for h in history:
87         counts[sum(h)] += 1
88
89     total = len(history)
90     probs = [counts[k]/total for k in range(5)]
91
92     plt.figure()
93     plt.bar(range(5), probs)
94     plt.xlabel("k = number of players playing 1")
95     plt.ylabel("Probability")
96     plt.title(title)
97     plt.show()
98
99 # Run experiments (multiple seeds)
100 def run_experiment(runs=5):
101     mwu_dists = []
102     swap_dists = []
103
104     for s in range(runs):
105         hist_mwu = run_mwu(s)
106         hist_swap = run_swap(s)
107
108         mwu_dists.append(get_distribution(hist_mwu))
109         swap_dists.append(get_distribution(hist_swap))
110
111     avg_mwu = np.mean(mwu_dists, axis=0)
112     avg_swap = np.mean(swap_dists, axis=0)
113

```

```

114     return avg_mwu, avg_swap, hist_mwu, hist_swap
115
116 # Execute
117 avg_mwu, avg_swap, hist_mwu, hist_swap = run_experiment()
118
119 # Profile distributions
120 plot_profile_dist(avg_mwu, "MWU (No External Regret)")
121 plot_profile_dist(avg_swap, "Swap Regret (Regret Matching)")
122
123 # k distributions (VERY IMPORTANT)
124 plot_k_dist(hist_mwu, "MWU: Distribution over k")
125 plot_k_dist(hist_swap, "Swap Regret: Distribution over k")

```

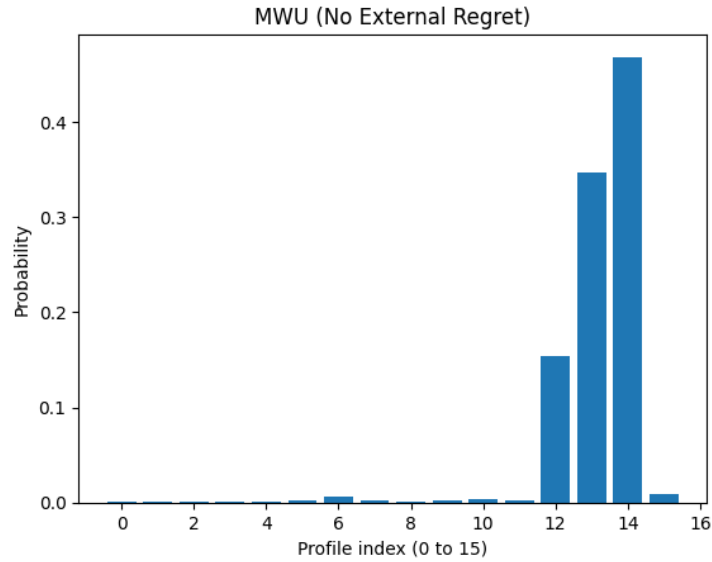


Figure 5: MWU: empirical distribution over action profiles

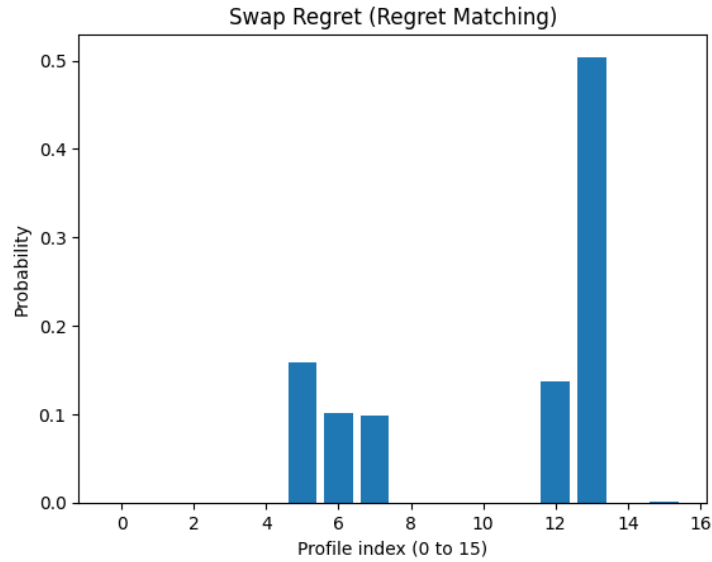


Figure 6: Swap regret: empirical distribution over action profiles

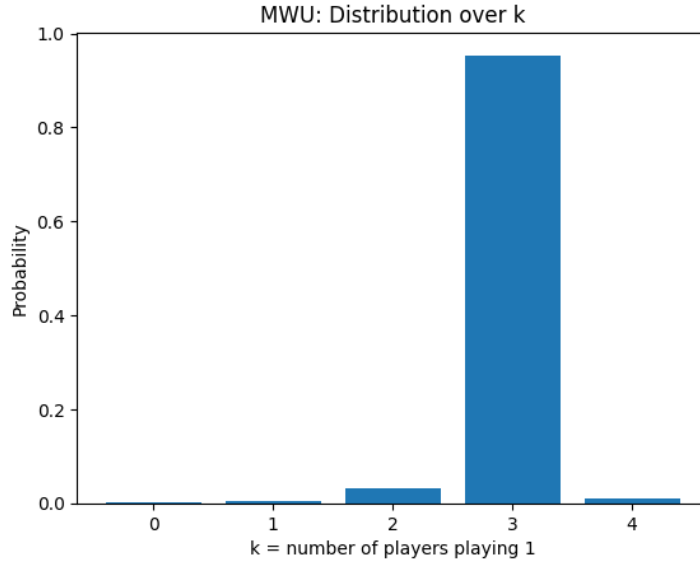


Figure 7: MWU: distribution over k

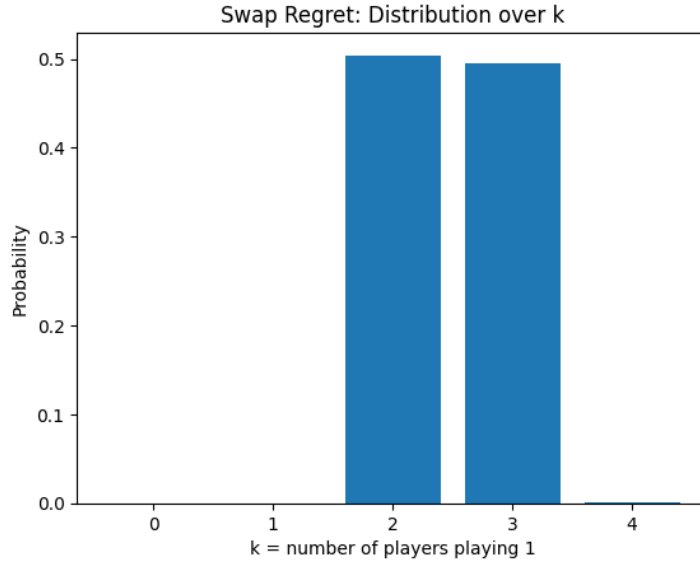


Figure 8: Swap regret: distribution over k

Observation:

Under MWU (no-external regret), the empirical distribution places most of the mass on profiles with $k = 3$, indicating convergence to a coarse correlated equilibrium.

Under regret matching (no-swap regret), the empirical distribution spreads over profiles with $k = 2$ and $k = 3$. This reflects convergence to the set of correlated equilibria, rather than a single equilibrium.

Profiles with $k=2$ correspond to correlated equilibria, while $k=3$ corresponds to coarse correlated equilibria.

The empirical distribution under MWU satisfies the coarse correlated equilibrium conditions due to vanishing external regret, while regret matching satisfies the stronger swap regret condition, implying convergence to a correlated equilibrium.

(v) From the simulations:

- Under no-external regret learning (MWU), the empirical distribution concentrates on profiles with $k = 3$, consistent with convergence to a coarse correlated equilibrium.

- Under no-swap regret learning (regret matching), the empirical distribution places mass on profiles with both $k = 2$ and $k = 3$. This is consistent with convergence to the set of correlated equilibria, which may include mixtures over multiple action profiles.